

Problem

A balanced delta-connected load is supplied by a 60-Hz three-phase source with a line voltage of 480 V. Each load phase draws 24 kW at a lagging power factor of 0.8. Find:

(a) the load impedance per phase

(b) the line current

(c) the value of capacitance needed to be connected in parallel with each load phase to minimize the current from the source

Step-by-step solution

Step 1 of 7

Calculate the value of the power factor angle θ .

$$\theta = \cos^{-1}(\text{pf})$$

Substitute 0.8 for pf.

$$\begin{aligned}\theta &= \cos^{-1}(0.8) \\ &= 36.87^\circ\end{aligned}$$

Therefore, the value of the power factor angle is, 36.87° .

Step 2 of 7

Calculate the value of the apparent power S .

$$S = \frac{P}{\text{pf}}$$

Substitute 24 kW for P and 0.8 for pf.

$$\begin{aligned}S &= \frac{24 \text{ k}}{0.8} \\ &= 30 \text{ kW}\end{aligned}$$

Therefore, the value of the apparent power S is, 30 kW.

Step 3 of 7

Calculate the value of the reactive power Q .

$$Q = S \sin \theta$$

Substitute 36.87° for θ and 30 kW for S .

$$\begin{aligned}Q &= (30 \text{ k})(\sin 36.87^\circ) \\ &= (30 \text{ k})(0.6) \\ &= 18 \text{ kVAR}\end{aligned}$$

Therefore, the value of the reactive power is Q is, 18 kVAR.

Step 4 of 7

Consider the expression for the complex power S .

$$S = P + jQ$$

Substitute 18kVAR for Q and 24 kW for P .

$$S = (24 \text{ k} + j18 \text{ k}) \text{ VA}$$

Therefore, the value of the apparent power S is, $(24 \text{ k} + j18 \text{ k}) \text{ VA}$.

Step 5 of 7

(a)

Consider the expression for the complex power

$$S = \frac{3(V_p)^2}{Z^*}$$

$$Z^* = \frac{3(V_p)^2}{S}$$

$$Z = \left[\frac{3(V_p)^2}{S} \right]^*$$

Substitute $(24 \text{ k} + j18 \text{ k}) \text{ VA}$ for S and 480 V for V_p .

$$\begin{aligned} Z &= \left[\frac{3(480)^2}{24 \text{ k} + j18 \text{ k}} \right]^* \\ &= \left[\frac{3(230400)}{24 \text{ k} + j18 \text{ k}} \right]^* \\ &= \left[\frac{691200}{24 \text{ k} + j18 \text{ k}} \right]^* \\ &= [18.432 - j13.824]^* \end{aligned}$$

$$Z = 18.432 + j13.824$$

Here Z represents the total value of the load impedance. To calculate the impedance per phase of the delta load, total impedance is to be divided by 3. That is,

$$\begin{aligned} Z_{ph} &= \frac{18.432 + j13.824}{3} \\ &= 6.144 + j4.608 \Omega \end{aligned}$$

Therefore, the value of the impedance per phase Z_{ph} is, $6.144 + j4.608 \Omega$.

[Comments \(2\)](#)



Anonymous

how did you do this calculation before the conjugate?



Anonymous

nvmmm i got it

Step 6 of 7

(b)

Calculate the value of the line current I_L .

$$I_L = \frac{P}{\sqrt{3}V_L(\text{pf})}$$

Substitute 24 kW for P , 0.8 for pf and 480 V for V_p .

$$\begin{aligned} I_L &= \frac{24 \text{ k}}{\sqrt{3}(480)(0.8)} \\ &= \frac{24 \text{ k}}{665.11} \\ &= 36.08 \text{ A} \end{aligned}$$

Therefore, the value of the line current I_L is, 36.08 A.

Step 7 of 7

(c)

For the current to minimize from the source, the value of the reactive power should be suppressed to zero by inserting a capacitor across the load.

Calculate the value of the capacitor that produces reactive power of 4.608 VAR.

$$C = \frac{Q}{2\pi fV^2}$$

Substitute 18 kVAR for Q , 60 Hz for f and 480 V_{rms} for V .

$$\begin{aligned} C &= \frac{18 \text{ k}}{2\pi(60)(480)^2} \\ &= \frac{18 \text{ k}}{2\pi(60)(230400)} \\ &= 207.2 \text{ }\mu\text{F} \end{aligned}$$

Therefore, the required value of the capacitor C to operate the circuit at unity power factor is,

207.2 μF .